

Knowledge Representation & Planning

Logical Reasoning and Forward Chaining

1. Logical Reasoning in AI

Logical reasoning is the process of deriving new facts from known facts using formal logic rules.

In Artificial Intelligence:

- Knowledge is represented using logical form.
- Inference rules are applied.
- New conclusions are generated automatically.

Goal: To enable machines to reason logically like humans.

2. Types of Logic Used in AI

2.1 Propositional Logic

- Statements are either True or False.
- Uses symbols like P, Q, R.
- Logical operators:
AND ($\wedge$), OR ($\vee$), NOT ($\neg$), IMPLIES ($\rightarrow$)

Example:

P: It is raining

Q: Roads are wet

Rule: $P \rightarrow Q$

If P is true, then Q must be true.

2.2 Predicate Logic (First Order Logic – FOL)

More expressive than propositional logic.

Uses:

- Predicates: Human(x), Student(x)
- Variables: x, y
- Quantifiers:
 - $\forall$ (for all)
 - $\exists$ (there exists)

Example:

$\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

Human(Socrates)

Conclusion:

Mortal(Socrates)

3. Logical Reasoning

Logical reasoning means applying inference rules to derive conclusions from a knowledge base.

Two main reasoning techniques:

1. Forward Chaining (Data-driven)
2. Backward Chaining (Goal-driven)

4. Forward Chaining

Definition:

Forward chaining is a data-driven inference technique that starts from known facts and applies rules to generate new facts until a goal is reached.

It is commonly used in:

- Rule-based expert systems
- Knowledge-based systems
- Production systems

5. Forward Chaining Algorithm

Step 1: Start with known facts.

Step 2: Check rules whose premises match the facts.

Step 3: Fire the rule.

Step 4: Add the conclusion to the fact base.

Step 5: Repeat until:

- Goal is reached, or
- No new facts can be derived.

6. Example 1 – Propositional Logic

Knowledge Base:

R1: $A \rightarrow B$

R2: $B \rightarrow C$

R3: $C \rightarrow D$

Fact:

A is true

Forward chaining:

- A is true
- Apply R1 $\rightarrow$ B becomes true
- Apply R2 $\rightarrow$ C becomes true
- Apply R3 $\rightarrow$ D becomes true

Final conclusion:

D is true

7. Example 2 – Medical Diagnosis Example

Knowledge Base:

R1: $\text{Fever} \wedge \text{Cough} \rightarrow \text{Flu}$

R2: $\text{Flu} \rightarrow \text{Rest}$

Facts:

Fever = True

Cough = True

Forward chaining:

- Fever and Cough satisfy R1
- Infer Flu
- Flu satisfies R2
- Infer Rest

Conclusion:

Patient should take Rest.

8. Example 3 – First Order Logic

Knowledge:

$\forall x \text{ Student}(x) \rightarrow \text{EligibleForExam}(x)$
 $\text{Student}(\text{Rahul})$

Forward chaining:

- $\text{Student}(\text{Rahul})$
- Apply rule
- $\text{EligibleForExam}(\text{Rahul})$

9. Pseudocode for Forward Chaining

While new facts can be inferred:

 For each rule:

 If premises match known facts:

 Add conclusion to fact base

10. Forward Chaining in AI and ML

Forward chaining is used in:

1. Expert Systems (medical diagnosis, fraud detection)
2. Production Rule Systems
3. Chatbots
4. Business Rule Engines
5. Decision tree-like reasoning systems

Although Machine Learning is data-driven, logical reasoning is important in:

- Explainable AI
- Knowledge Graphs
- Hybrid Neuro-Symbolic AI systems

11. Forward vs Backward Chaining

Forward Chaining:

- Data-driven
- Starts from facts
- Good when many facts are available

Backward Chaining:

- Goal-driven
- Starts from query
- Good for diagnosis systems

12. Advantages of Forward Chaining

- Simple and easy to implement
- Suitable for real-time systems
- Useful when facts continuously arrive

13. Disadvantages

- May generate unnecessary facts
- Can be slow with large rule bases
- Not efficient when goal is specific

14. Backward Chaining

Definition:

Backward chaining is a goal-driven inference technique that starts from a goal (query) and works backward to determine whether known facts support that goal.

Instead of starting from facts, it starts from what we want to prove.

15. Working of Backward Chaining

Step 1: Start with the goal (query).

Step 2: Find a rule whose conclusion matches the goal.

Step 3: Replace the goal with the rule's premises (sub-goals).

Step 4: Check whether sub-goals are true in the fact base.

Step 5: Repeat until:

- All sub-goals are satisfied, or
- No rule can support the goal.

16. Example 1 – Propositional Logic

Knowledge Base:

R1: $A \rightarrow B$

R2: $B \rightarrow C$

R3: $C \rightarrow D$

Facts:

A is true

Goal: Prove D

Backward chaining process:

- Goal: D
- Find rule R3 ($C \rightarrow D$)
New sub-goal: C
- Find rule R2 ($B \rightarrow C$)
New sub-goal: B
- Find rule R1 ($A \rightarrow B$)
New sub-goal: A
- A is true (fact found)

Since A is true,

B becomes true $\rightarrow$ C becomes true $\rightarrow$ D becomes true

Goal proved.

17. Example 2 – Medical Diagnosis

Knowledge Base:

R1: Fever $\wedge$ Cough $\rightarrow$ Flu

R2: Flu $\rightarrow$ Rest

Facts:

Fever = True

Cough = True

Goal: Should patient take Rest?

Backward chaining steps:

- Goal: Rest
- Find rule R2 (Flu $\rightarrow$ Rest)
Sub-goal: Flu
- Find rule R1 (Fever $\wedge$ Cough $\rightarrow$ Flu)
Sub-goals: Fever and Cough
- Both are true

Therefore:

Flu is true $\rightarrow$ Rest is true

Goal achieved.

18. Example 3 – First Order Logic

Knowledge:

$\forall x \text{ Student}(x) \rightarrow \text{EligibleForExam}(x)$

Student(Rahul)

Goal: EligibleForExam(Rahul)

Backward chaining:

- Goal: EligibleForExam(Rahul)
- Use rule: Student(x) $\rightarrow$ EligibleForExam(x)
- Sub-goal: Student(Rahul)
- Found in facts

Therefore, goal is proved.

19. Pseudocode for Backward Chaining

Function BackwardChaining(goal):

 If goal is in facts:

 Return True

 Find rules whose conclusion matches goal

 For each rule:

 Try to prove all premises (sub-goals)

 If all premises are true:

 Return True

 Return False

20. Forward vs Backward Chaining (Comparison)

Forward Chaining:

- Data-driven
- Starts from known facts
- May generate many intermediate facts
- Useful in monitoring systems

Backward Chaining:

- Goal-driven
- Starts from query
- Focused search
- Efficient when goal is specific

21. Applications of Backward Chaining in AI

1. Expert systems (medical diagnosis systems like MYCIN)
2. Prolog programming language
3. Question-answering systems
4. Automated theorem proving
5. Planning systems

22. Backward Chaining in Modern AI and ML

Although deep learning dominates AI:

- Used in Explainable AI systems
- Used in rule-based reasoning engines
- Used in knowledge graph query systems
- Used in logic programming (Prolog)

It is important in Hybrid AI (Neuro-Symbolic AI), where logical reasoning is combined with machine learning.

23. Advantages of Backward Chaining

- Efficient when goal is known
- Reduces unnecessary fact generation
- Focused reasoning
- Suitable for diagnostic systems

24. Disadvantages

- Not suitable when multiple goals must be derived
- Can be complex if many rules match a goal
- Requires recursive reasoning